The Complexity–Spuriousness Trade-off in Probing: Toward a Unified Capacity Measure for Diagnostic and Causal Interpretability

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Abstract

Diagnostic probing and causal probing (via distributed alignment search, DAS) are supervised interpretability techniques, both of which tend to find what one looks for: (i) a sufficiently powerful diagnostic probe always recovers a target property from a model's representations (Pimentel et al., 2020); and a sufficiently complex causal probe always recovers an alignment between any algorithm and a neural network (Sutter et al., 2025). Both these statements, however, are about a limit: at infinite complexity, either method can fit any target, including random ones, so their reported scores say nothing about a neural network. Read at face value, these results say that causal probing is no better than diagnostic probing—it leaves us where diagnostic probing did. But we only know this is true at the infinite complexity limit. In practice, we do not work at that limit, and at finite complexity and finite data, the extra structure in a causal probe might make it more resistant to fitting *spurious* targets. This report proposes measuring that directly—how well a method fits randomly relabelled data, which is a version of its empirical Rademacher complexity—in a form that applies unchanged to diagnostic probing and to DAS, and asks how this spurious-fitting capacity grows with complexity for each. We describe the measure and an experimental protocol on the hierarchical equality task, and report preliminary results: on the correct labels, both a linear probe and a linear DAS map recover the task's intermediate variable, so the feature is at once decodable and interveneable. A first sweep over dataset size finds DAS fitting random relabellings no more readily than the probe—and clearly less so at the most informative size—while matching it on signal: a preliminary, single-seed sign that causal probing is the safer of the two at finite scale.

1 Motivation

A probe is a supervised model trained to predict some property T from a network's representations R; if the probe succeeds, we are tempted to conclude that the representations *encode* the property (Alain and Bengio, 2017; Hewitt and Manning, 2019; Belinkov, 2022). Pimentel et al. (2020) recast this practice information-theoretically: probing estimates the mutual information $\mathrm{I}(T; R)$, and the best estimate is obtained by the *best-performing* probe one can train, regardless of its complexity. The corollary is uncomfortable: under their Section 2 assumptions, a contextual representation contains exactly as much information about a word-level property as the underlying sentence does, so a powerful enough probe recovers the property for reasons that need have nothing to do with how—or whether—the network uses it.

Causal probing was meant to escape this critique by intervening rather than merely decoding. Distributed alignment search (DAS; Geiger et al., 2024) learns a (possibly non-linear) alignment map ϕ onto the hidden space, performs interchange interventions in the aligned basis, and measures the *interchange-intervention accuracy* (IIA) with which the network reproduces a target algorithm's counterfactual behaviour. Sutter et al. (2025) show that this escape is illusory: once the linearity constraint on ϕ is lifted, *any* algorithm can be aligned to *any* network with near-perfect IIA—even randomly initialised networks that cannot solve the task. They call this the **non-linear representation dilemma**: lifting the linearity constraint removes any principled way to trade off the alignment map's complexity against its accuracy.

Read together, the two results are one failure seen twice: in each, an interpretability method reports a score—probe accuracy, or IIA—chosen as the best fit over a family of extraction functions, and in each, that score stops being informative once the family is expressive enough. The practitioner has no principled way to set the family's complexity.

There is a pessimistic reading of this, and it is worth stating plainly. Sutter et al. (2025) show that in the limit of unconstrained alignment maps, causal abstraction is vacuous in the same way diagnostic probing is: the method succeeds regardless of what the network computes. At face value, this says that moving from diagnostic to causal probing buys us nothing—we are left where Pimentel et al. (2020) left us. But it is a claim about the limit, and it need not hold in practice. At the finite map complexities people actually use, the extra structure in a causal probe—it must act through the frozen network, and stay consistent across several interventions—might make it harder to fit spurious patterns than a diagnostic probe of comparable power. If so, causality does add something: not in the limit, but in the regime we work in. Whether it does is an empirical question. To answer it we need a measure of how readily each method fits spurious targets that applies to both—a single notion of capacity, in the spirit of Rademacher complexity (Bartlett and Mendelson, 2002) or VC dimension (Vapnik, 1998)—and a way to watch it grow with complexity.

2 Setup and notation

We adopt notation compatible with the two companion papers. A network induces a representation map $x \mapsto R$, with $R \in \mathbb{R}^d$. An **interpretability method** $\mathcal{M}$ is characterised by three ingredients: (i) a *hypothesis class* $\mathcal{H}_c$ indexed by a scalar *complexity knob* $c \in \mathbb{R}_+$; (ii) a per-example *fit* functional $\mathrm{fit}(h; \cdot) \in [0, 1]$ measuring how well a hypothesis $h \in \mathcal{H}_c$ reproduces a target on one example; and (iii) a *selection rule* that, given a finite sample $\mathcal{D}_n$ of size n, returns $\hat{h} = \arg\max_{h \in \mathcal{H}_c} \widehat{\mathrm{fit}}(h; \mathcal{D}_n)$ and reports the achieved fit $s(\mathcal{M}) = \widehat{\mathrm{fit}}(\hat{h}; \mathcal{D}_n)$.

The two methods instantiate this template as follows.

Diagnostic probing. $\mathcal{H}_c$ is a family of probes $q_{\boldsymbol{\theta}}(t \mid \mathbf{r})$—e.g. multi-layer perceptrons whose depth or width grows with c, as in Pimentel et al. (2020). The targets are property labels t, the inputs are activations $\mathbf{r}$, and $\widehat{\mathrm{fit}}$ is accuracy (or, in the information-theoretic view, the negative cross-entropy that lower-bounds $\mathrm{I}(T; R)$). The dataset is $\mathcal{D}_n = \{(\mathbf{r}_i, t_i)\}_{i=1}^n$.

Causal probing (DAS). $\mathcal{H}_c$ is a family of bijective alignment maps ϕ—e.g. reversible networks (Gomez et al., 2017) whose depth L_{rn} or hidden size d_{rn} grows with c, as in Sutter et al. (2025). The "label" of an example is the counterfactual output the *target algorithm* prescribes, the inputs are (base, source) pairs, and $\widehat{\mathrm{fit}}$ is the IIA: the rate at which the *frozen* network, intervened upon in the aligned basis, reproduces that output. The dataset is $\mathcal{D}_n = \{(\mathbf{x}_i^{\mathrm{base}}, \mathbf{x}_i^{\mathrm{src}}, \mathbf{y}_i^{\mathrm{cf}})\}_{i=1}^n$.

The crucial structural difference—which we will argue is the heart of the matter—is that for DAS the hypothesis is not a classifier of $\mathbf{r}$. It is a *reparametrisation* ϕ of the hidden space, after which a *fixed* intervention and the *fixed* downstream network produce the prediction. The induced labelling function is $x \mapsto \mathrm{N}\big(\mathrm{intervene}_\phi(x)\big)$, and the family is constrained to invertible ϕ.

3 A unified capacity measure

3.1 Spuriousness as fit to randomised targets

We want to measure how much of a method's reported fit could be spurious—how well it fits targets that carry no real signal about the network. The direct way to do this is to relabel the data at random and check how well the method still fits it. The expected best fit a hypothesis class achieves on such random targets is, up to scaling, its empirical *Rademacher complexity* (Bartlett and Mendelson, 2002); we use it, in accuracy space, as our measure of spurious-fitting capacity.

Definition 1 (Spurious-fitting capacity)**.** *Fix a representation sample of size n and let $\boldsymbol{\sigma}$ denote a target* randomisation*: a relabelling of the examples that destroys any genuine relationship between inputs and targets while preserving their marginals. The spurious-fitting capacity of a method $\mathcal{M}$ at complexity c is*

$$\mathfrak{R}_n(\mathcal{H}_c) \;\coloneqq\; \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{h\in\mathcal{H}_c} \widehat{\mathrm{fit}}\big(h;\, \boldsymbol{\sigma}(\mathcal{D}_n)\big)\right] \;-\; s_{\mathrm{chance}}, \tag{1}$$

i.e. the expected best fit the class achieves on randomised targets, centred at chance performance.

This quantity is, up to a scaling constant and the usual move from a margin loss to 0/1 accuracy, the empirical Rademacher complexity of $\mathcal{H}_c$ on the representation sample. Its appeal is threefold. First, it is *method-agnostic*: it specialises to diagnostic probing (random token labels) and to DAS (a random target algorithm, or equivalently permuted counterfactual outputs) without changing form. Second, it is exactly the quantity practitioners already estimate: it is the control-task accuracy of Hewitt and Liang (2019), and the random-network alignment of Sutter et al. (2025) is its causal analogue.[1] Third, it is *cheap*: a Monte-Carlo estimate requires only re-running the existing fitting pipeline on permuted targets.

3.2 The limit and the finite regime

Definition 1 is not what the two theorems are about, and the difference matters. The theorems are about a limit: at infinite complexity, both a probe and an alignment map can fit any target on any amount of data, random targets included (Pimentel et al., 2020; Sutter et al., 2025, Thm. 1). In that limit the two methods are equally uninformative and nothing tells them apart.

We never run at that limit. We fit at finite complexity on finite data, where neither method can fit everything. There, the two can differ: a linear probe and a linear rotation do not have the same ability to fit random relabellings, and that ability grows differently as we make each more complex. Definition 1 measures exactly this ability, and its growth with complexity is what we study—not whether the methods break in the limit (they do), but how their capacity to fit spurious targets grows on the way there.

[1]We deliberately phrase Definition 1 in accuracy space rather than as the textbook $\mathbb{E}_{\boldsymbol{\sigma}}[\sup_h \frac{1}{n}\sum_i \sigma_i h(\mathbf{r}_i)]$ so that it is directly comparable across the two methods and directly estimable from the existing code paths, which already train on randomised labels.

3.3 What "graceful" should mean, and the comparability problem

A natural temptation is to compare the two methods by plotting their capacity against a shared notion of "size"—say, parameter count—and asking whether one grows like $\mathcal{O}(\sqrt{P})$ where the other grows like $\mathcal{O}(P)$. We caution against taking the exponent itself as the contribution, for two reasons. First, generic norm- or parameter-based bounds yield the *same* $\mathcal{O}(\sqrt{P/n})$ form for any reasonable class (Neyshabur et al., 2015), so an a priori difference in exponent is not expected; a measured difference would have to come from structure, not from counting. Second, "depth of an MLP probe" and "depth of a RevNet alignment map" are not the same unit, so a raw c-axis is not commensurable.

Both problems are solved by using *capacity itself* as the common axis. Concretely, we propose to plot, for each complexity level, the **signal** (fit on *correct* targets) against the **spurious capacity** (fit on *randomised* targets, Definition 1). Each method traces a curve in this *signal–spurious plane* as c varies. A method **fails gracefully** if its curve hugs the top-left—high signal recovered per unit of spurious capacity spent—and **fails sharply** if signal and spurious capacity rise together. This operationalises "one method fails more gracefully than the other" without ever committing to incommensurable units, and it directly generalises the selectivity of Hewitt and Liang (2019) (which is the single-point gap signal − spurious) into a *scaling curve*.

4 A mechanism for divergence

If the two methods do diverge in the signal–spurious plane, the explanation must lie in structural constraints that one method imposes and the other does not. We identify three constraints that are present in DAS but absent in diagnostic probing, each of which should *throttle* spurious capacity below what the raw parameter count of ϕ would suggest:

1. **Invertibility.** The alignment map ϕ is constrained to be a bijection. A bijection cannot collapse or discard directions of the hidden space to manufacture a fit; it can only rotate and warp them. This is a genuine capacity restriction with no analogue in an unconstrained probe.
2. **The frozen network.** The prediction is produced by the *fixed* downstream network applied to an intervened, ϕ-aligned hidden state. The map cannot directly emit a label; it can only reroute information that the frozen weights will then process. Spurious fits must survive this fixed computation.
3. **Multi-intervention consistency.** A single ϕ must simultaneously satisfy the counterfactual constraints of *several* intervention types (in the hierarchical equality task, the two sub-equalities and their conjunction). These are many coupled constraints, not one label, and jointly satisfying them on randomised targets is harder than fitting a single randomised label.

Our working hypothesis is that these constraints make $\mathfrak{R}_n^{\mathrm{DAS}}(c)$ grow *more slowly* than $\mathfrak{R}_n^{\mathrm{DP}}(c)$—i.e. that DAS, despite the alarming $c \to \infty$ result of Sutter et al. (2025), may be the more graceful failure at the finite capacities used in practice. We stress that this is a hypothesis to be tested, not a claim; the constraints above could equally be swamped by the sheer flexibility a non-linear ϕ buys, in which case DAS would fail *more* sharply. Either outcome is informative.

5 Experimental protocol

We instantiate the study on the **hierarchical equality task**, the toy setting of Sutter et al. (2025), for which an implementation already exists in this repository (a 3-layer MLP trained to decide whether two pairs of vectors are "both equal or both unequal", together with linear-probe and linear-DAS pipelines and *randomised-label* controls). The task is ideal because the ground-truth intermediate variables (WX, YZ) are known, so "signal" is unambiguous.

Complexity axes. We add a complexity knob to each method. For *diagnostic probing*, we replace the single logistic-regression probe with a family of MLP probes of increasing depth and width ($c \in \{$linear, 1-hidden-layer, 2-hidden-layer, $\dots\}$), as in Pimentel et al. (2020). For *DAS*, we replace the orthogonal (rotation) alignment map with a family of RevNet maps of increasing depth L_{rn} and hidden size d_{rn}, as in Sutter et al. (2025). The data budget n is held *fixed* across the sweep so that we isolate the effect of c (in contrast to the existing sweeps, which vary n).

Measurements. At each complexity level and for several seeds we record (i) the *signal*: fit on correct targets (probe accuracy / IIA on the trained network); and (ii) the *spurious capacity* $\widehat{\mathfrak{R}}_n(\mathcal{H}_c)$ of Definition 1, estimated by re-fitting on randomised targets (random token labels for probing; a randomly relabelled / random target algorithm for DAS). We additionally repeat the DAS measurement on a *randomly initialised* network, which makes the spurious axis vivid: any IIA there is, by construction, spurious.

Analysis. We (1) plot signal and spurious capacity against c for each method; (2) plot the two methods together in the *signal–spurious plane* of Section 3; and (3) fit a simple growth model $\widehat{\mathfrak{R}}_n(c) \approx a\, c^b$ to read off, and compare, the empirical growth exponents b with confidence intervals. The deliverable is a statement of the form "per unit of spurious capacity, method X recovers more genuine signal than method Y on this task," with the supporting curves.

What would falsify the framing. If signal and spurious capacity rise in lockstep for *both* methods (curves collapsing onto the diagonal of the signal–spurious plane), then capacity carries no method-distinguishing information and the unification, while tidy, would be practically empty. We regard making this falsifiable as a feature.

6 Preliminary results

We report initial measurements on the hierarchical equality task. The signal-side results below are stable; the spurious-capacity comparison—the report's central quantity—comes from the scaling sweep, which we are still running.

WX is linearly decodable, and linearly interveneable. A linear probe on the trained MLP's first-layer activations recovers WX with about 0.98 accuracy (decodability decreases with depth; the output O becomes near-perfectly decodable only in the deeper layers, $0.70 \to 0.997$). A linear (rotation) alignment map trained at that same layer to perform the WX interchange reaches about 0.999 interchange-intervention accuracy. So, on the correct labels, both methods recover WX, and recover it strongly: the feature is both decodable by a probe and interveneable by a linear map.

An earlier negative result was a setup artefact. Before restricting to a single intervention, we measured DAS accuracy near chance and spent some effort on it. That was not a property of linear alignment. The counterfactual generator produced a multi-intervention mix in which the joint (both-subspace) intervention yielded label-homogeneous batches—no training signal—and a single shared rotation was being asked to satisfy several intervention objectives at once. Restricting to the

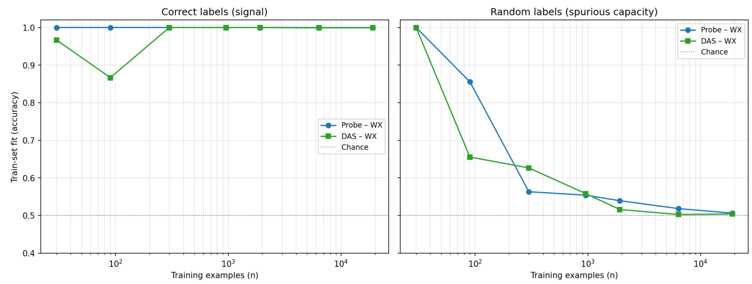

Figure 1: Train-set fit vs. number of training examples n, for the diagnostic probe and linear DAS, on correct labels (signal, solid) and a fixed random relabelling (spurious capacity, dashed). Both recover WX on the correct labels; on random labels both fall from perfect memorisation at small n toward chance, and DAS's curve lies at or below the probe's at most sizes (most clearly at $n = 90$), indicating lower spurious-fitting capacity. Single seed.

single WX intervention (balanced batches, one objective) makes the rotation converge cleanly: its gradient vanishes and the learned rotation stabilises as accuracy approaches 1. We flag this because the failure was instructive—with causal probing it is easy to mistake a data or optimisation artefact for a substantive "the model does not encode this causally" conclusion.

Spurious-fitting capacity. Figure 1 runs the sweep of Section 5: train-set fit against n, on the correct labels and on a fixed random relabelling, for both methods. On the correct labels both fit WX essentially perfectly across the range (the probe at 1.0; DAS at ≈ 1.0 for $n \gtrsim 300$). On random labels both start at 1.0 at the smallest n—any method can memorise 30 points—and both decay toward chance by $n \approx 2 \times 10^4$. In between, where capacity rather than data is the binding constraint, DAS's random-label fit sits at or below the probe's at most sizes, most clearly at $n = 90$ (probe ≈ 0.86, DAS ≈ 0.66). Taken at face value, DAS reaches the same signal while fitting noise less readily—its spurious-fitting capacity is the smaller of the two—which is the direction the finite-complexity hypothesis predicts. We report this as suggestive, not settled: the curves are single-seed and noisy (they cross once, at $n = 300$, and DAS's correct-label fit wobbles at the smallest n), so the gap should be confirmed by averaging over seeds before it is leaned on.

What we did not do. Two extensions would turn this into a clean result: averaging over seeds (with error bands) to confirm the gap, and sweeping a genuine *complexity* axis (deeper probes; RevNet alignment maps) rather than dataset size. We leave both to future work.

7 Implementation notes

A few methodological points, mostly explaining why the setup looks the way it does.

Why a single intervention. We restrict to the WX interchange. With the joint (both-subspace) intervention the data generator produces label-homogeneous batches—each block fixes both intermediate values, so the output is constant within a block—which carry no training signal; and a single shared rotation could not, in our hands, satisfy the single and joint interventions together. Both problems disappear with a single, balanced intervention, which is what let DAS converge (Section 6).

Judge causal probes by held-out, order-independent evaluation. The obvious training-time metric—a running average of interchange accuracy—is misleading. The counterfactual data is stored in contiguous single-type blocks, and a running average over them drifts with the order the blocks arrive in, which can look exactly like a model that "learns then collapses" even when the alignment map is frozen. Every number we report comes from a single order-independent pass over a held-out set.

A pyvene gotcha. The fast intervention path keeps only the first of several source locations; with the (now removed) two-source joint intervention this silently dropped the second source. With a single source it is correct and much faster, so we use it.

8 Relation to prior work, and a caveat on consistency

The control-task literature is the obvious antecedent. Hewitt and Liang (2019) introduce *selectivity*—task accuracy minus control-task accuracy—as a criterion for choosing probes, and Voita and Titov (2020) propose minimum description length as a complexity-aware alternative. Our spurious-fitting capacity is, at a single complexity level, exactly a control-task score; the novelty here is (i) reading it as a Rademacher complexity, (ii) lifting it from diagnostic to *causal* probing, and (iii) studying its *scaling* rather than its value at one point.

We must, however, confront a tension with Pimentel et al. (2020), which argues *against* selectivity as a probe-*selection* criterion: penalising a complex probe for its control-task accuracy artificially discards capacity that yields a tighter information estimate. We do not retract that argument; we repurpose the quantity. Selectivity is a poor *objective* to optimise—one should still pick the best-performing probe for estimation—but it is an excellent *descriptive diagnostic* of how much a method *could* fool itself, and its growth rate is a property of the method, not a knob to tune. Measuring fragility is not the same as selecting against it.

Finally, a caveat on what Definition 1 measures. It is about fitting random targets at finite data and finite complexity. This is related to, but not the same as, what the two theorems prove—that at infinite complexity any target can be fit. A method could be reluctant to fit random targets at the complexities we actually test and still be able to fit anything in the limit. Our measure speaks to the former, the regime we use in practice, not the latter.

9 Conclusion

We have argued that the diagnostic critique of Pimentel et al. (2020) and the causal critique of Sutter et al. (2025) are two instances of the same thing: a method whose reported fit grows with its capacity to fit anything, spurious targets included. We proposed measuring that capacity directly—how well a method fits randomly relabelled data—in a form that applies to both diagnostic probing and DAS, and studying how it grows with complexity. Our preliminary results on the hierarchical equality task show that, on the correct labels, the task's intermediate variable is both linearly decodable by a probe ($\sim$0.98) and linearly interveneable by an alignment map ($\sim$0.999 IIA); an earlier near-chance DAS reading turned out to be an artefact of a multi-intervention data and optimisation setup, not a property of linear alignment. With both recovering the signal, the comparison that matters is spurious-fitting capacity, and a first (single-seed) sweep over dataset size points the predicted way: DAS fits random labels no more readily than the probe, and less readily at the most informative size. If this survives seed-averaging and a genuine complexity axis, it is a concrete sense in which causal probing is the safer tool at finite complexity—even though, in the limit, it is no better than the diagnostic probing it was meant to improve on.

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